

WMDR2

Package in package 'Model'

WMDR2

Version 0.3.1 Phase 1.0 Proposed

jkl created on 2026-03-06. Last modified 2026-07-09

WMDR2 diagram

Class diagram in package 'WMDR2'

WMDR2 v0.3.x

v0.3.0

- Cardinality of facility.deployments changed to [0..*]
- Deployment.location [0..1] added to describe relative location of instrument as free text.
- Deployment.serialNumber moved to ObservingConfiguration.InstrumentInstance.serialNumber
- Deployment.instrument associated to InstrumentInstance
- Deployment.sourceOfObservation moved to ObservingConfiguration
- As a consequence of the above, Deployment describes the location context only and is renamed to ObservingLocation.
- "reporting" folded into "ReportingProcedure"
- ReportingProcedure.validFrom dropped.
- Reporting.spatialReportingInterval added.
- cardinality contact.organization from [0..1] to 1
- cardinality wmoRegion from [0..1] to 1
- cardinality of ReportingProcedure.dataPolicy from [0..1] to 1; nilReason allowed
- cardinality of ReportingProcedure.internationalExchange from [0..1] to 1
- cardinality of ObservingConfiguration.operatingStatus from [0..1] to 1
- cardinality of ObservingConfiguration.sourceOfObservation from [0..1] to 1

v0.3.1

- remove ObservingLocation, move all attributes to ObservingConfiguration.
- rename validFrom to time, which is specified as an OGC period / interval.
- move serialNumber directly under observingConfiguration, drop separate instrumentInstance.
- observingConfiguration.operatingStatus changed from 1 to 0..1

WMDR2

Version 0.3.1

jkl created on 2026-03-06. Last modified 2026-07-10

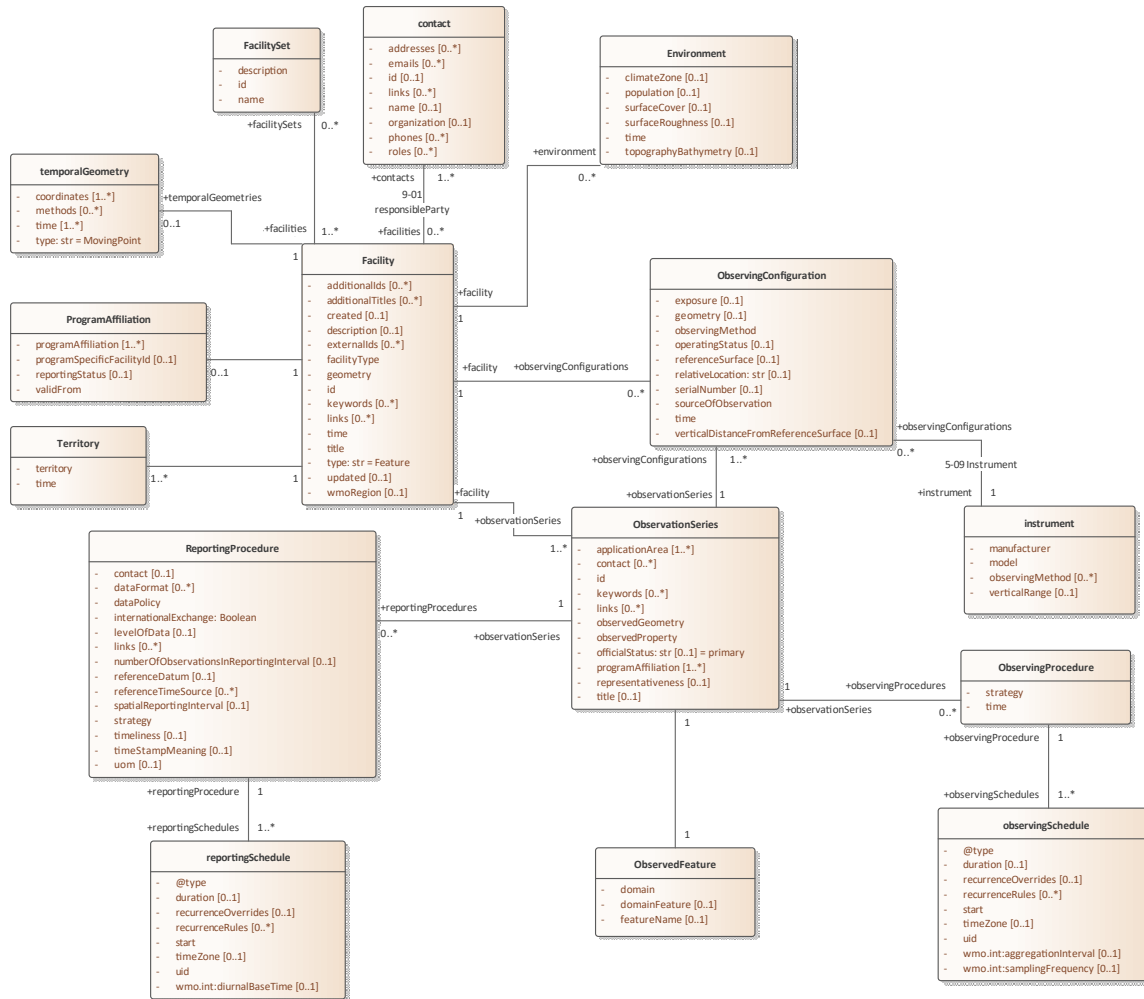


Figure 1: WMDR2

contact

Class in package 'WMDR2'

WMDR2 core JSON class: contact.

contact
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-11. Last modified 2026-06-25

ATTRIBUTES

◆ addresses : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Postal addresses.

[Is static True. Containment is Not Specified.]

◆ emails : Private

ATTRIBUTES

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Email contacts.

[Is static True. Containment is Not Specified.]



id : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Optional stable contact identifier.

[Is static True. Containment is Not Specified.]



links : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Online resources associated with the contact.

[Is static True. Containment is Not Specified.]



name : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Either a full name, or a sequence of name parts. Examples:

Dr. Jörg Klausen (this corresponds to FN in vcard)

Klausen;Jörg;Dr.; (this corresponds to N in vcard)

[Is static True. Containment is Not Specified.]



organization : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Example:

ORG:Federal Office of Meteorology and Climatology MeteoSwiss

[Is static True. Containment is Not Specified.]



phones : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Phone contacts.

[Is static True. Containment is Not Specified.]



roles : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Specific contact roles such as owner, pointOfContact, principalInvestigator.

[Is static True. Containment is Not Specified.]

ASSOCIATIONS



Association (direction: Unspecified) 9-01 responsibleParty

Source: Public facilities (Class) Facility

Cardinality: [0..*]

Target: Public contacts (Class) contact

Cardinality: [1..*]

Environment

Class in package 'WMDR2'

Facility-environment object containing climate, surface-cover, population, and topography/bathymetry histories.

Environment

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-02. Last modified 2026-06-25

ATTRIBUTES	
 climateZone : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 population : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Population in 10 and 50 km radius around the facility	[Is static True. Containment is Not Specified.]
 surfaceCover : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 surfaceRoughness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 time : Private conformant to OGC time: [beginDate, endDate]	[Is static True. Containment is Not Specified.]
 topographyBathymetry : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public facility (Class) Facility Cardinality: [1]	Target: Public environment (Class) Environment Cardinality: [0..*]

Extension

Class in package 'WMDR2'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

Extension
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-11. Last modified 2026-06-26

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public describes (Class) ObservationSeries Cardinality: [1..*]	Target: Public uses (Class) Extension Cardinality: [0..1] ObservingProcedure(s)

Facility

Class in package 'WMDR2'

Facility properties payload of the WMDR2 full record. Root Features geometry/time are modeled on fullRecord, not here.

Facility
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-06-25

ATTRIBUTES	
 additionalIds : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Additional WSIs for the facility.	[Is static True. Containment is Not Specified.]
 additionalTitles : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Additional names for the facility, i.e., "aliases" that may be in use.	[Is static True. Containment is Not Specified.]
 created : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Record creation date/time metadata.	[Is static True. Containment is Not Specified.]
 description : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Facility description.	[Is static True. Containment is Not Specified.]
 externalIds : Private Alias: 3-06 (primary & additional) WSIs	

ATTRIBUTES	
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) External identifiers such as WMO:WIGOS.	[Is static True. Containment is Not Specified.]
 facilityType : Private Alias: 3-04 facilityType Compact WMDR facility type code value.	[Is static True. Containment is Not Specified.]
 geometry : Private Alias: 3-07 current geolocation geometry = geolocation = coordinates + elevation	[Is static True. Containment is Not Specified.]
 id : Private Alias: 3-06 (primary) WSI The primary WIGOS Station Identifier	[Is static True. Containment is Not Specified.]
 keywords : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Discovery/search keywords retained in the core JSON.	[Is static True. Containment is Not Specified.]
 links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Related links.	[Is static True. Containment is Not Specified.]
 time : Private Alias: [dateEstablished, dateClosed] mandatory element in OGC Records API Part 1, an interval. Used to declare [dateEstablished, dateClosed]	[Is static True. Containment is Not Specified.]
 title : Private Alias: 3-03 facilityName properties.title = 3-03 facilityName	[Is static True. Containment is Not Specified.]
 type : str Private = Feature Facility properties type; const facility.	[Is static True. Containment is Not Specified.]

ATTRIBUTES

updated : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Record update date/time metadata.

[Is static True. Containment is Not Specified.]

wmoRegion : Private
 Alias: 3-01 wmoRegion
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

mandatory for fixed facilities, optional for mobile facilities.

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public facilities (Class) Facility
 Cardinality: [1..*]

Target: Public facilitySets (Class) FacilitySet
 Cardinality: [0..*]

Association (direction: Unspecified)

Source: Public facility (Class) Facility
 Cardinality: [1]

Target: Public observingLocations (Class)
 ObservingLocation
 Cardinality: [0..*]

Association (direction: Unspecified) 9-01 responsibleParty

Source: Public facilities (Class) Facility
 Cardinality: [0..*]

Target: Public contacts (Class) contact
 Cardinality: [1..*]

Association (direction: Unspecified)

Source: Public facility (Class) Facility
 Cardinality: [1]

Target: Public temporalGeometries (Class)
 temporalGeometry
 Cardinality: [0..1]

Association (direction: Unspecified)

Source: Public facility (Class) Facility
 Cardinality: [1]

Target: Public environment (Class) Environment
 Cardinality: [0..*]

Association (direction: Unspecified)

Source: Public (Class) Facility
 Cardinality: [1]

Target: Public (Class) Territory
 Cardinality: [1..*]

Association (direction: Unspecified)

Source: Public facility (Class) Facility

Target: Public observationSeries (Class)

ASSOCIATIONS	
Cardinality: [1]	ObservationSeries Cardinality: [1..*]
 Association (direction: Unspecified)	
Source: Public (Class) Facility Cardinality: [1]	Target: Public (Class) ProgramAffiliation Cardinality: [0..1]
 Association (direction: Unspecified)	
Source: Public observingConfigurations (Class) ObservingConfiguration Cardinality: [0..*]	Target: Public facility (Class) Facility Cardinality: [1]

FacilitySet

Class in package 'WMDR2'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

FacilitySet
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-06-25

ATTRIBUTES	
 description : Private	[Is static True. Containment is Not Specified.]
 id : Private	[Is static True. Containment is Not Specified.]
 name : Private Alias: title	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public facilities (Class) Facility Cardinality: [1..*]	Target: Public facilitySets (Class) FacilitySet Cardinality: [0..*]

instrument

Class in package 'WMDR2'

Describes instrument capabilities, not a specific instrument instance.

For semantic consistency, terms like `observedProperty` and `observedGeometry` are used, even though they just describe the instrument capabilities, not the actual observations. With this, instrument becomes re-usable between deployments and even facilities.

instrument
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-05-22. Last modified 2026-06-25

ATTRIBUTES	
 manufacturer : Private Alias: 5-09 manufacturer	[Is static True. Containment is Not Specified.]
 model : Private Alias: 5-09 model	[Is static True. Containment is Not Specified.]
 observingMethod : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 verticalRange : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Specify using an array of [lower boundary, upper boundary] in meter. This is a capability of an instrument as specified, not the actually accessible vertical range.	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public groups (Class) instrument Cardinality: [0..1]	Target: Public belongs_to (Class) instrumentSet Cardinality: [0..1]
 Association (direction: Unspecified) 5-09 Instrument Source: Public observingConfigurations (Class) ObservingConfiguration Cardinality: [0..*]	Target: Public instrument (Class) instrument Cardinality: [1]

InstrumentInstance

Class in package 'WMDR2'

InstrumentInstance
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-07-06. Last modified 2026-07-06

ASSOCIATIONS
 Association (direction: Unspecified)

ASSOCIATIONS

Source: Public instrumentInstance (Class) ObservingConfiguration
Cardinality: [0..1]

Target: Public observingConfiguration (Class)
InstrumentInstance
Cardinality: [0..1]

instrumentSet

Class in package 'WMDR2'

instrumentSet

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-06-22. Last modified 2026-06-25

ATTRIBUTES

 description : Private

[Is static True. Containment is Not Specified.]

 id : Private

[Is static True. Containment is Not Specified.]

 name : Private
Alias: title

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public groups (Class) instrument
Cardinality: [0..1]

Target: Public belongs_to (Class) instrumentSet
Cardinality: [0..1]

ObservationSeries

Class in package 'WMDR2'

WMDR2 core JSON class: observation.

ObservationSeries

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-06. Last modified 2026-06-25

ATTRIBUTES

 applicationArea : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Value(s) from <http://codes.wmo.int/wmdr/ApplicationArea>

[Is static True. Containment is Not Specified.]

ATTRIBUTES

◆ contact : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Observation-level contacts.

[Is static True. Containment is Not Specified.]

◆ id : Private

Observation identifier, e.g. observation:<observedVariable>.

[Is static True. Containment is Not Specified.]

◆ keywords : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Observation keywords.

[Is static True. Containment is Not Specified.]

◆ links : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Observation links.

[Is static True. Containment is Not Specified.]

◆ observedGeometry : Private

cf. <http://codes.wmo.int/wmdr/Geometry>

[Is static True. Containment is Not Specified.]

◆ observedProperty : Private
Alias: OMS:ObservedProperty, iop:Variable

The observedVariable is constructed following I-ADOPT ontology approaches, with elements described in a unified vocabulary hosted at codes.wmo.int/wmdr2.

NB: observedVariable will be a LONG list of entries, each record comprised of notation, name, description. The notation is a unique identifier, not necessarily human-readable. The name is potentially human-readable. The description is a fully fledged definition, human-readable and describing all relevant aspects of an observed variable. It is derived from the observedProperty, the object of interest, the matrix and domain, as well as constraints. Matrix and constraints are defined in controlled vocabularies that may be domain-specific.

Examples include:

- observableProperty: state variable (e.g., temperature, pressure, leaf-area-index (LAI), ...), or another property (e.g., thickness, concentration, scattering coefficient, color, ...)
- objectOfInterest: sea ice, CO2, aerosol particles, chlorophyll
- observablePropertyConstraints: a conventional level above ground (e.g., 2m above ground for temperature measurements), a size fraction (e.g., PM2.5), a specific wavelength (e.g., 635 nm), a pressure (range), a temperature (range), a humidity (range), etc., as relevant for the result of the observation. Constraints shall be documented as name-value pairs, where the names shall come from a code list, and the values shall come from a code list depending on the name, or be free text.
- matrix: atmosphere, interstitial air in the root space, air in an ice core, sea surface, soil, dry air, ...

These elements will be supported by code lists from codes.wmo.int/wmdr2.

Applications may want to guide users to select a specific observedVariable by offering groupings, filtering, or tree-like visualizations according to matrix, objectOfInterest, etc.

[Is static True. Containment is Not Specified.]

◆ officialStatus : str Private = primary

ATTRIBUTES

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Values from <http://codes.wmo.int/wmdr/ObservationStatus>

Defaults to "primary" if not provided.

[Is static True. Containment is Not Specified.]

 **programAffiliation** : Private

Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Program-affiliation timeline fragments inherited from WMDR1 observations.

[Is static True. Containment is Not Specified.]

 **representativeness** : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

<http://codes.wmo.int/wmdr/Representativeness>

[Is static True. Containment is Not Specified.]

 **title** : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Human-readable title derived from domain, geometry, observed variable, and optional label.

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public describes (Class) ObservationSeries
Cardinality: [1..*]

Target: Public uses (Class) Extension
Cardinality: [0..1]

ObservingProcedure(s)

 Association (direction: Unspecified)

Source: Public describes (Class) ObservationSeries
Cardinality: [1]

Target: Public has (Class) OfficialStatus
Cardinality: [0..1]

 Association (direction: Unspecified)

Source: Public observationSeries (Class) ObservationSeries
Cardinality: [1]

Target: Public observingProcedures (Class)
ObservingProcedure
Cardinality: [0..*]

 Association (direction: Unspecified)

Source: Public observationSeries (Class) ObservationSeries
Cardinality: [1]

Target: Public reportingProcedures (Class)
ReportingProcedure
Cardinality: [0..*]

 Association (direction: Unspecified)

ASSOCIATIONS	
Source: Public (Class) ObservationSeries	Target: Public (Class) ReferenceSurface
 Association (direction: Unspecified)	
Source: Public (Class) ObservationSeries Cardinality: [1]	Target: Public (Class) ObservedFeature Cardinality: [1]
 Association (direction: Unspecified) 3-07 temporalGeometry	
Source: Public (Class) ObservationSeries Cardinality: [1]	Target: Public (Class) temporalGeometry Cardinality: [0..1]
describes	has_geolocatino
 Association (direction: Unspecified)	
Source: Public groups (Class) ObservationSeries Cardinality: [1..*]	Target: Public belongs_to (Class) ObservationSeriesSet Cardinality: [0..*]
 Association (direction: Unspecified)	
Source: Public observationSeries (Class) ObservationSeries Cardinality: [1]	Target: Public observingConfigurations (Class) ObservingConfiguration Cardinality: [1..*]
 Association (direction: Unspecified)	
Source: Public facility (Class) Facility Cardinality: [1]	Target: Public observationSeries (Class) ObservationSeries Cardinality: [1..*]

ObservationSeriesSet

Class in package 'WMDR2'

grouping of observations (which are themselves time series)

ObservationSeriesSet
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-06-25

ATTRIBUTES	
 description : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
Deployment temporal extent, date-resolution interval or null.	[Is static True. Containment is Not Specified.]

ATTRIBUTES

 id : Private

Deployment identifier; source XML id is preserved when available.

[Is static True. Containment is Not Specified.]

 name : Private

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public groups (Class) ObservationSeries
Cardinality: [1..*]

Target: Public belongs_to (Class)
ObservationSeriesSet
Cardinality: [0..*]

ObservedFeature

Class in package 'WMDR2'

Describes the physical domain, i.e. larger-scale environmental target or ultimate-feature-of-interest, of an observation.

ObservedFeature
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-06-10. Last modified 2026-06-25

ATTRIBUTES

 domain : Private

The largest scale in WIGOS, cf. <http://codes.wmo.int/wmdr/Domain>

[Is static True. Containment is Not Specified.]

 domainFeature : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Describes the type of the larger-scale environmental target or ultimate-feature-of-interest, of an observation.

Concept motivated by requirements from hydrology, but may be useful elsewhere. New code list is proposed, with terms like

- river basin
- lake
- river
- catchment

- canopy

- epilimnion
- hypolimnion

- troposphere
- tropopause
- stratosphere

- thermocline (useful as a general term, but needs "domain")

ATTRIBUTES

- boundary layer (useful as a general term, but needs "domain")

- ...

[Is static True. Containment is Not Specified.]

◆ featureName : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

free text field, e.g., "Danube Basin", "Amazon rain forest", "Amazon river Basin"

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

✎ Association (direction: Unspecified)

Source: Public (Class) ObservationSeries

Cardinality: [1]

Target: Public (Class) ObservedFeature

Cardinality: [1]

ObservingConfiguration

Class in package 'WMDR2'

ObservingConfiguration

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-06-26. Last modified 2026-07-06

ATTRIBUTES

◆ exposure : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Values from <http://codes.wmo.int/wmdr/Exposure>

[Is static True. Containment is Not Specified.]

◆ geometry : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

geometry is the OGC Records name for geolocation.

For stationary facilities, this represents the geolocation of the sensor / inlet, and should be documented. If not specified, the geometry (and temporalGeometry) of the facility is assumed. For mobile facilities, this should be left blank.

[Is static True. Containment is Not Specified.]

◆ observingMethod : Private

from (new) unified code list of observing methods, based on

<http://codes.wmo.int/wmdr/ObservingMethodAtmosphere>

<http://codes.wmo.int/wmdr/ObservingMethodTerrestrial>

<http://codes.wmo.int/wmdr/ObservingMethodEarth>

[Is static True. Containment is Not Specified.]

◆ operatingStatus : Private

ATTRIBUTES

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

values refer to <http://codes.wmo.int/wmdr/InstrumentOperatingStatus>

[Is static True. Containment is Not Specified.]

referenceSurface : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Linked to heightAboveReferenceSurface, from <http://codes.wmo.int/wmdr/ReferenceSurfaceType>

[Is static True. Containment is Not Specified.]

relativeLocation : str Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Free text, documenting the relative location of instrument / sensor / inlet / observer with respect to the geometry (and temporalGeometry) of the facility.

[Is static True. Containment is Not Specified.]

serialNumber : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]

sourceOfObservation : Private

Value from <http://codes.wmo.int/wmdr/SourceOfObservation>.

If sourceOfObservation=="humanObservation", several attributes of "observations" are expected to be null.

[Is static True. Containment is Not Specified.]

time : Private

conformant to OGC time: [beginDate, endDate]

[Is static True. Containment is Not Specified.]

verticalDistanceFromReferenceSurface : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

implement as an array of objects:

{"value": float, "uom": str}, whereby "uom" from <http://codes.wmo.int/wmdr/unit>

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public observingConfigurations (Class)
ObservingConfiguration
Cardinality: [0..*]

Target: Public facility (Class) Facility
Cardinality: [1]

Association (direction: Unspecified) 5-09 Instrument

ASSOCIATIONS	
Source: Public observingConfigurations (Class) ObservingConfiguration Cardinality: [0..*]	Target: Public instrument (Class) instrument Cardinality: [1]
 Association (direction: Unspecified)	
Source: Public instrumentInstance (Class) ObservingConfiguration Cardinality: [0..1]	Target: Public observingConfiguration (Class) InstrumentInstance Cardinality: [0..1]
 Association (direction: Unspecified)	
Source: Public observationSeries (Class) ObservationSeries Cardinality: [1]	Target: Public observingConfigurations (Class) ObservingConfiguration Cardinality: [1..*]

ObservingLocation

Class in package 'WMDR2'

Documents the placement of a particular instrument type (from a catalogue) at a specific geolocation (=geometry) or relativeLocation for the purpose of making measurements. Can also describe the position of a human observer.

ObservingLocation
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-06-22. Last modified 2026-07-06

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public facility (Class) Facility Cardinality: [1]	Target: Public observingLocations (Class) ObservingLocation Cardinality: [0..*]

ObservingProcedure

Class in package 'WMDR2'

When are observations made? Describes an observingStrategy, and depending on it, a temporal schedule/coverage, e.g. "from Feb-Oct on working days from 06-18 UTC".
Implement as array of objects

ObservingProcedure
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-11. Last modified 2026-06-25

ATTRIBUTES
 strategy : Private
Values from http://codes.wmo.int/wmdr/SamplingStrategy

ATTRIBUTES	
[Is static True. Containment is Not Specified.]	
 time : Private conformant to OGC time: [beginDate, endDate]	
[Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public observingProcedure (Class) ObservingProcedure Cardinality: [1]	Target: Public observingSchedules (Class) observingSchedule Cardinality: [1..*]
 Association (direction: Unspecified)	
Source: Public observationSeries (Class) ObservationSeries Cardinality: [1]	Target: Public observingProcedures (Class) ObservingProcedure Cardinality: [0..*]

observingSchedule

Class in package 'WMDR2'

Reusable JSCalendar Event schedule used by deployment.temporalObservingSchedule references. WMDR-specific extensions are represented with wmo.int:* properties.

May be expected for strategy "continuous" or "discrete"

observingSchedule

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-01. Last modified 2026-06-25

ATTRIBUTES	
 @type : Private JSCalendar object type; const Event.	
[Is static True. Containment is Not Specified.]	
 duration : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Occurrence duration describing coverage window or sampling duration.	
[Is static True. Containment is Not Specified.]	
 recurrenceOverrides : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) JSCalendar recurrence override patches; null override values are invalid.	
[Is static True. Containment is Not Specified.]	

ATTRIBUTES

🔑 recurrenceRules : Private
 Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

JSCalendar recurrence rules.

[Is static True. Containment is Not Specified.]

🔑 start : Private

Canonical JSCalendar recurrence anchor; date 0001-01-01 with time from coverage start.

[Is static True. Containment is Not Specified.]

🔑 timeZone : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

IANA time zone or UTC default.

[Is static True. Containment is Not Specified.]

🔑 uid : Private

JSCalendar-safe schedule uid.

[Is static True. Containment is Not Specified.]

🔑 wmo.int:aggregationInterval : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Describes the smallest interval over which raw/ level 0 observations are aggregated and stored.

Implement as {"value": int, "uom": str}

[Is static True. Containment is Not Specified.]

🔑 wmo.int:samplingFrequency : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

The frequency at which raw / level 0 observations are obtained.

Implement as {"value": int, "uom": str}

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

✎ Association (direction: Unspecified)

Source: Public observingProcedure (Class) ObservingProcedure
 Cardinality: [1]

Target: Public observingSchedules (Class)
 observingSchedule
 Cardinality: [1..*]

OfficialStatus

Class in package 'WMDR2'

OfficialStatus
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-06-25. Last modified 2026-06-25

ATTRIBUTES	
 date : Private	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public describes (Class) ObservationSeries Cardinality: [1]	Target: Public has (Class) OfficialStatus Cardinality: [0..1]

ProgramAffiliation

Class in package 'WMDR2'

ProgramAffiliation
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 2026-06-01. Last modified 2026-06-25
Alias temporalProgramAffiliation

ATTRIBUTES	
 programAffiliation : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 programSpecificFacilityId : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 reportingStatus : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) http://codes.wmo.int/wmdr/ReportingStatus	[Is static True. Containment is Not Specified.]
 validFrom : Private	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public (Class) Facility Cardinality: [1]	Target: Public (Class) ProgramAffiliation Cardinality: [0..1]

ReferenceSurface

Class in package 'WMDR2'

ReferenceSurface
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-06-26. Last modified 2026-06-26

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public (Class) ObservationSeries

Target: Public (Class) ReferenceSurface

reporting

Class in package 'WMDR2'

Reusable reporting template.

reporting
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 2026-06-01. Last modified 2026-06-23

ASSOCIATIONS

 Association (direction: Unspecified) 7-03 Temporal reporting

Source: Public procedure (Class) ReportingProcedure
Cardinality: [1]

Target: Public reporting (Class) reporting
Cardinality: [0..1]

ReportingProcedure

Class in package 'WMDR2'

ReportingProcedure
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-06-23. Last modified 2026-06-25

ATTRIBUTES

 contact : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]

 dataFormat : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Aligned array, values from <http://codes.wmo.int/wmdr/DataFormat>.

[Is static True. Containment is Not Specified.]

ATTRIBUTES	
◆ dataPolicy : Private Value from http://codes.wmo.int/wmdr/DataPolicy. nilReason acceptable.	[Is static True. Containment is Not Specified.]
◆ internationalExchange : Boolean Private Declaration if data are intended for international exchange or not.	[Is static True. Containment is Not Specified.]
◆ levelOfData : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Value from http://codes.wmo.int/wmdr/LevelOfData.	[Is static True. Containment is Not Specified.]
◆ links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) URL to related WIS record, entry point for data discovery and retrieval.	[Is static True. Containment is Not Specified.]
◆ numberOfObservationsInReportingInterval : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
◆ referenceDatum : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Value from http://codes.wmo.int/wmdr/CoordinateReferenceSystem.	[Is static True. Containment is Not Specified.]
◆ referenceTimeSource : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Single value or array, values from http://codes.wmo.int/wmdr/ReferenceTime.	[Is static True. Containment is Not Specified.]
◆ spatialReportingInterval : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) To be implemented as {"value": [float], "uom": str}, whereby len([float])<3 to report either a distance or a grid / pixel -dimension.	[Is static True. Containment is Not Specified.]
◆ strategy : Private	[Is static True. Containment is Not Specified.]

ATTRIBUTES

-  **timeliness** : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
 Expected timeliness of reporting, an object with value and uom (default: seconds)
 [Is static True. Containment is Not Specified.]
-  **timeStampMeaning** : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
 Aligned array, values from <http://codes.wmo.int/wmdr/TimeStampMeaning>.
 [Is static True. Containment is Not Specified.]
-  **uom** : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
 Values from <http://codes.wmo.int/wmdr/unit>.
 [Is static True. Containment is Not Specified.]

ASSOCIATIONS

-  Association (direction: Unspecified)
 Source: Public reportingProcedure (Class) ReportingProcedure
 Cardinality: [1]
 Target: Public reportingSchedules (Class) reportingSchedule
 Cardinality: [1..*]
-  Association (direction: Unspecified) 7-03 Temporal reporting
 Source: Public procedure (Class) ReportingProcedure
 Cardinality: [1]
 Target: Public reporting (Class) reporting
 Cardinality: [0..1]
-  Association (direction: Unspecified)
 Source: Public observationSeries (Class) ObservationSeries
 Cardinality: [1]
 Target: Public reportingProcedures (Class) ReportingProcedure
 Cardinality: [0..*]

reportingSchedule

Class in package 'WMDR2'

Reusable JSCalendar Event schedule used by deployment.temporalObservingSchedule references. WMDR-specific extensions are represented with wmo.int:* properties.

reportingSchedule
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 2026-06-25. Last modified 2026-06-25

ATTRIBUTES

-  **@type** : Private

ATTRIBUTES

JSCalendar object type; const Event.

[Is static True. Containment is Not Specified.]

duration : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Occurrence duration describing coverage window or sampling duration.

[Is static True. Containment is Not Specified.]

recurrenceOverrides : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

JSCalendar recurrence override patches; null override values are invalid.

[Is static True. Containment is Not Specified.]

recurrenceRules : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

JSCalendar recurrence rules.

To be used for "reportingInterval"; use start, duration and other elements to evaluate constraints if needed.

[Is static True. Containment is Not Specified.]

start : Private

Canonical JSCalendar recurrence anchor; date 0001-01-01 with time from coverage start.

[Is static True. Containment is Not Specified.]

timeZone : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

IANA time zone or UTC default.

[Is static True. Containment is Not Specified.]

uid : Private

JSCalendar-safe schedule uid.

[Is static True. Containment is Not Specified.]

wmo.int:diurnalBaseTime : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Relevant for 24h aggregates, e.g., precipitation amount collected over 24h.

Implement as {"time": hh:mm, "time_zone": str}

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified)

ASSOCIATIONS

Source: Public reportingProcedure (Class) ReportingProcedure
Cardinality: [1]

Target: Public reportingSchedules (Class)
reportingSchedule
Cardinality: [1..*]

temporalGeometry

Class in package 'WMDR2'

temporalGeometry

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-11. Last modified 2026-06-25

Alias temporalGeometry

ATTRIBUTES

 coordinates : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Coordinate history array; each entry is lon, lat[, elevation].

[Is static True. Containment is Not Specified.]

 methods : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

use <http://codes.wmo.int/wmdr/GeopositioningMethod>

[Is static True. Containment is Not Specified.]

 time : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Date array aligned with coordinates.
conformant to OGC time: [beginDate, endDate]

[Is static True. Containment is Not Specified.]

 type : str Private = MovingPoint

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public facility (Class) Facility
Cardinality: [1]

Target: Public temporalGeometries (Class)
temporalGeometry
Cardinality: [0..1]

temporalGeometry

Class in package 'WMDR2'

WMDR2 core JSON class: temporalGeometry.

temporalGeometry
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-06-12. Last modified 2026-06-02

ATTRIBUTES	
 coordinates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Coordinate history array; each entry is lon, lat[, elevation]. [Is static True. Containment is Not Specified.]	
 dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with coordinates. [Is static True. Containment is Not Specified.]	
 methods : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) use http://codes.wmo.int/wmdr/GeopositioningMethod [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
 Association (direction: Unspecified) 3-07 temporalGeometry Source: Public (Class) ObservationSeries Cardinality: [1] describes	Target: Public (Class) temporalGeometry Cardinality: [0..1] has_geolocatino

Territory

Class in package 'WMDR2'

WMDS reference: 3-02

mandatory for fixed facilities, optional for mobile facilities.

Implemented as [{"date": date, "territory": str}, {}] like other historical<Name> classes.

Territory
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-06-23. Last modified 2026-06-25

ATTRIBUTES	
 territory : Private Alias: temporal Territory Values from http://codes.wmo.int/wmdr/TerritoryName [Is static True. Containment is Not Specified.]	

ATTRIBUTES time : Private

conformant to OGC time: [beginDate, endDate]

[Is static True. Containment is Not Specified.]

ASSOCIATIONS Association (direction: Unspecified)Source: Public (Class) Facility
Cardinality: [1]Target: Public (Class) Territory
Cardinality: [1..*]